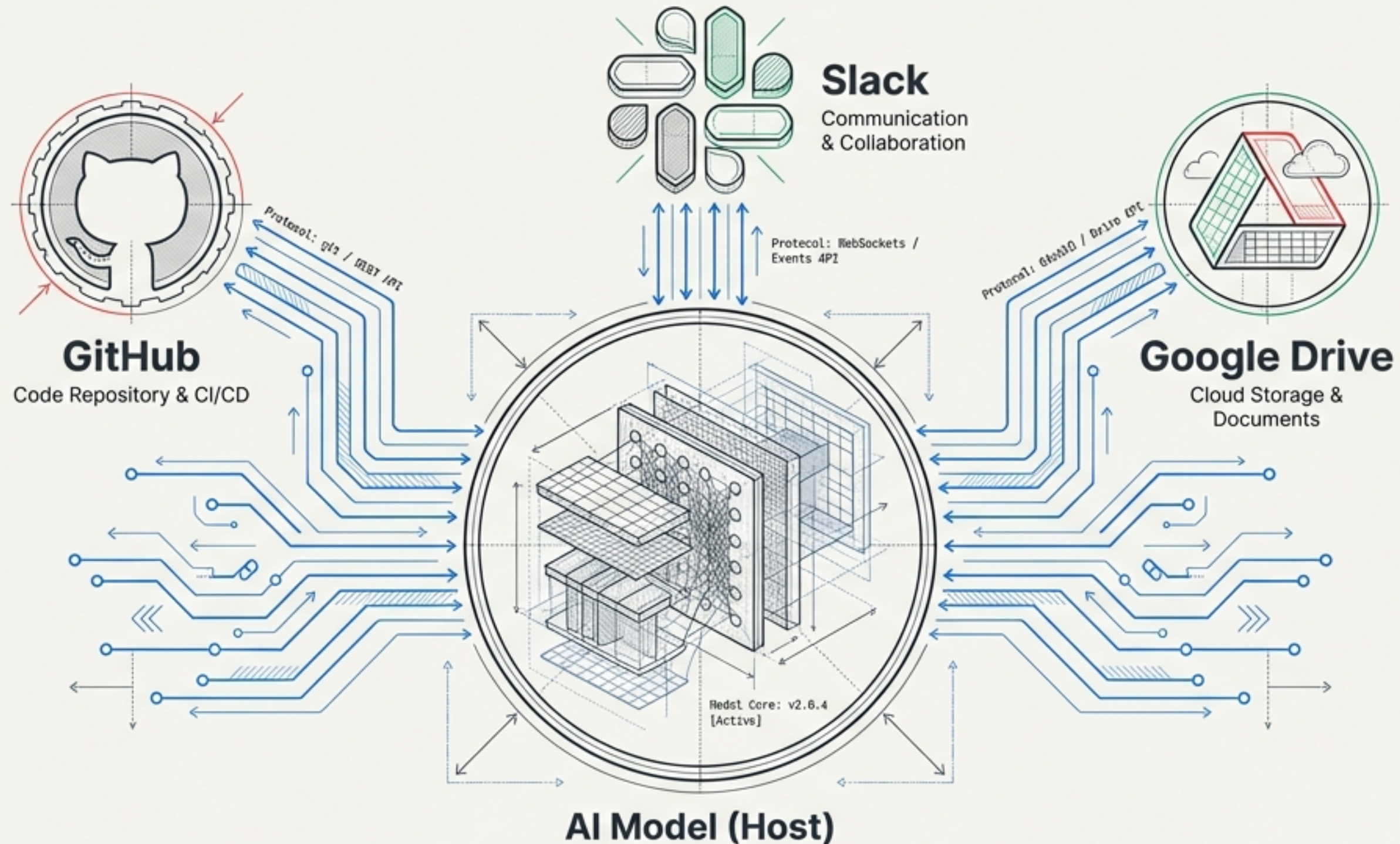


The Anatomy of MCP

A First-Principles Deep Dive into Connecting AI to the World



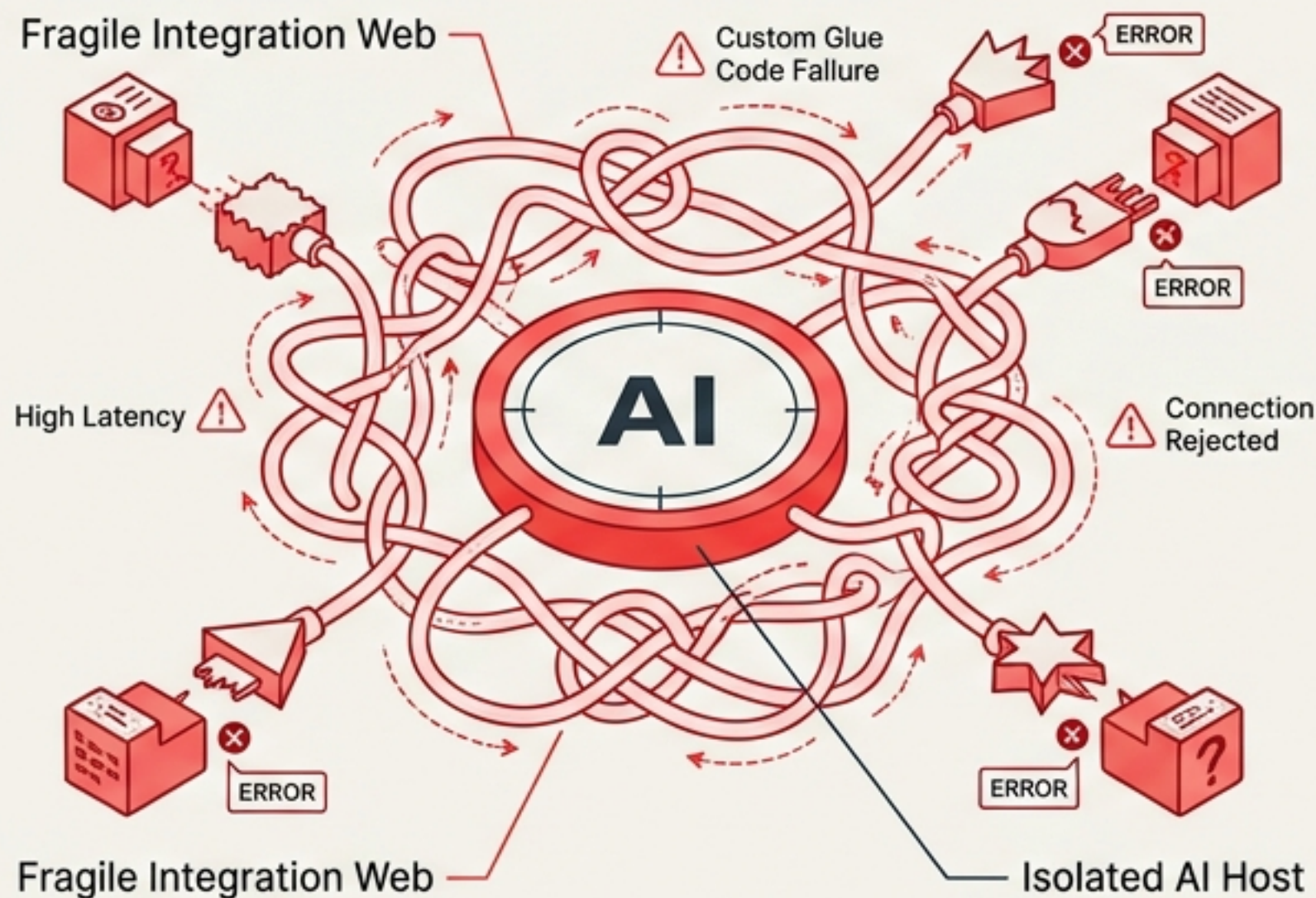
Based on the architectural breakdown by CampusX

The Universal Port for Intelligence

The Problem

High Friction & Fragility

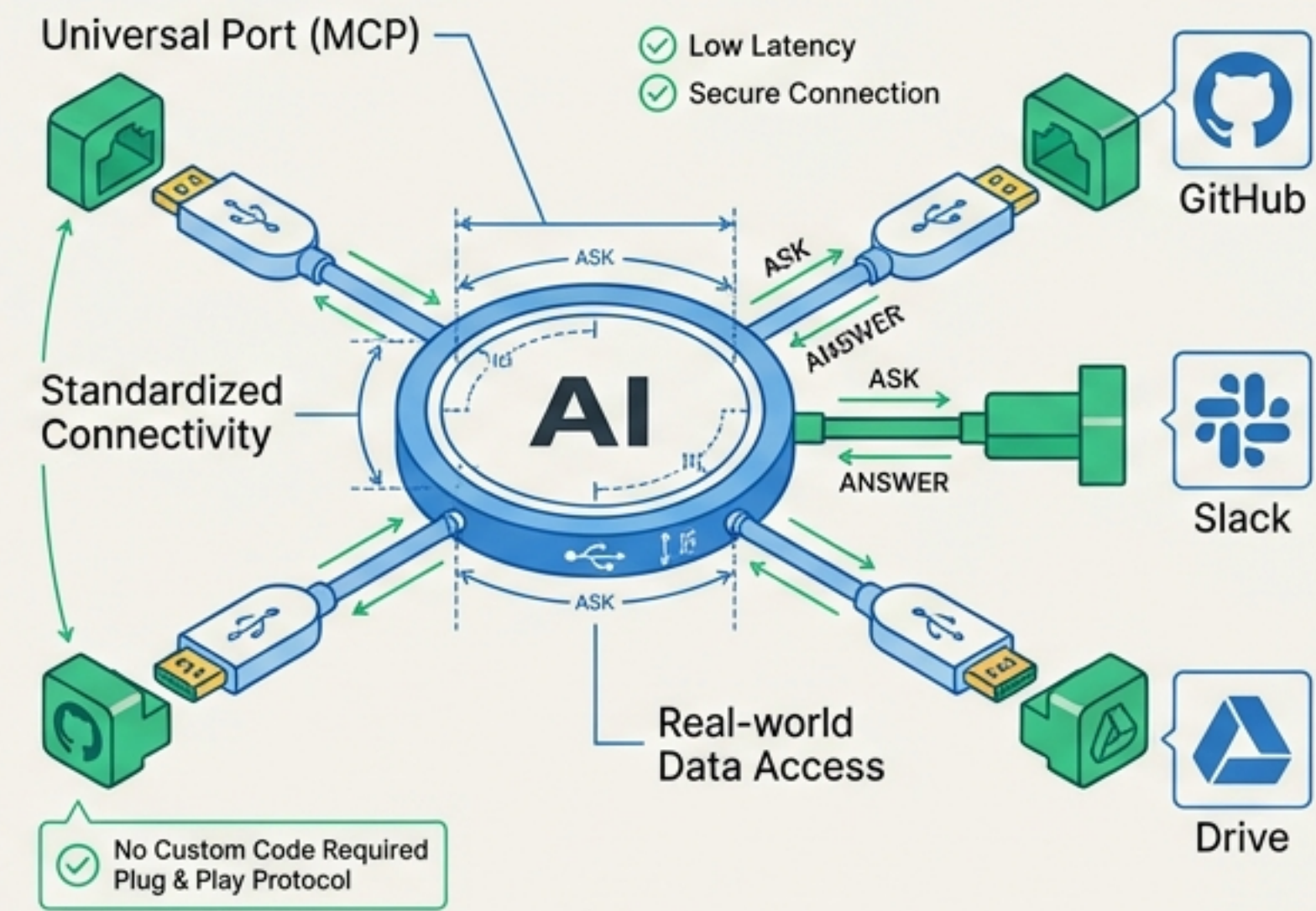
AI Hosts are isolated from real-world data. Custom API integrations create a fragile web of dependencies.



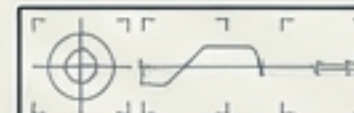
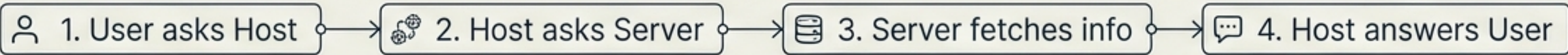
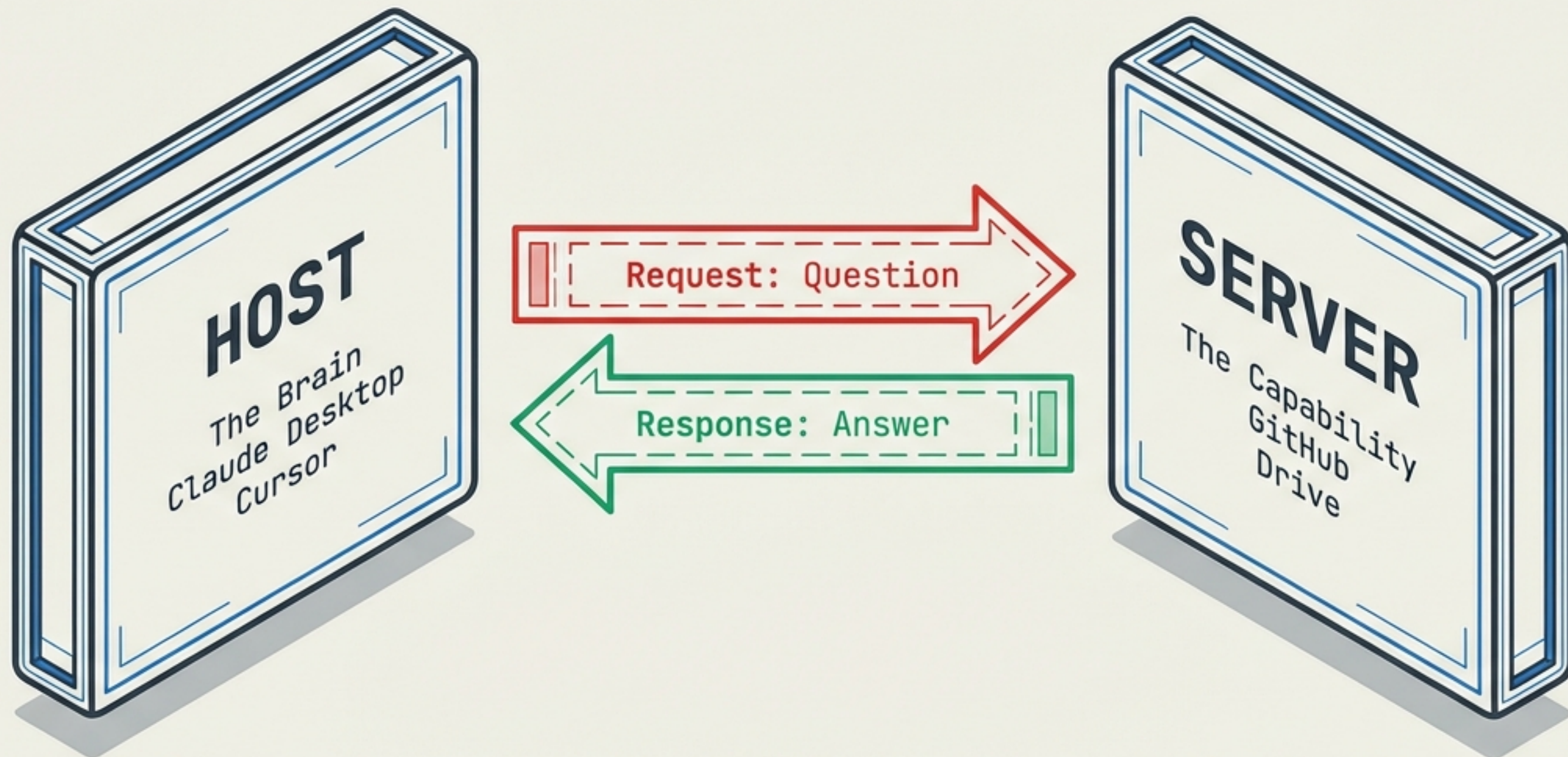
The Solution

Standardized Connectivity

MCP acts like a USB-C port for LLMs, enabling a universal "ask and answer" protocol without custom glue code.



The 10,000 ft View: Two Core Entities



TECHNICAL SCHEMATIC | VERSION 1.0

MODEL CONTEXT PROTOCOL

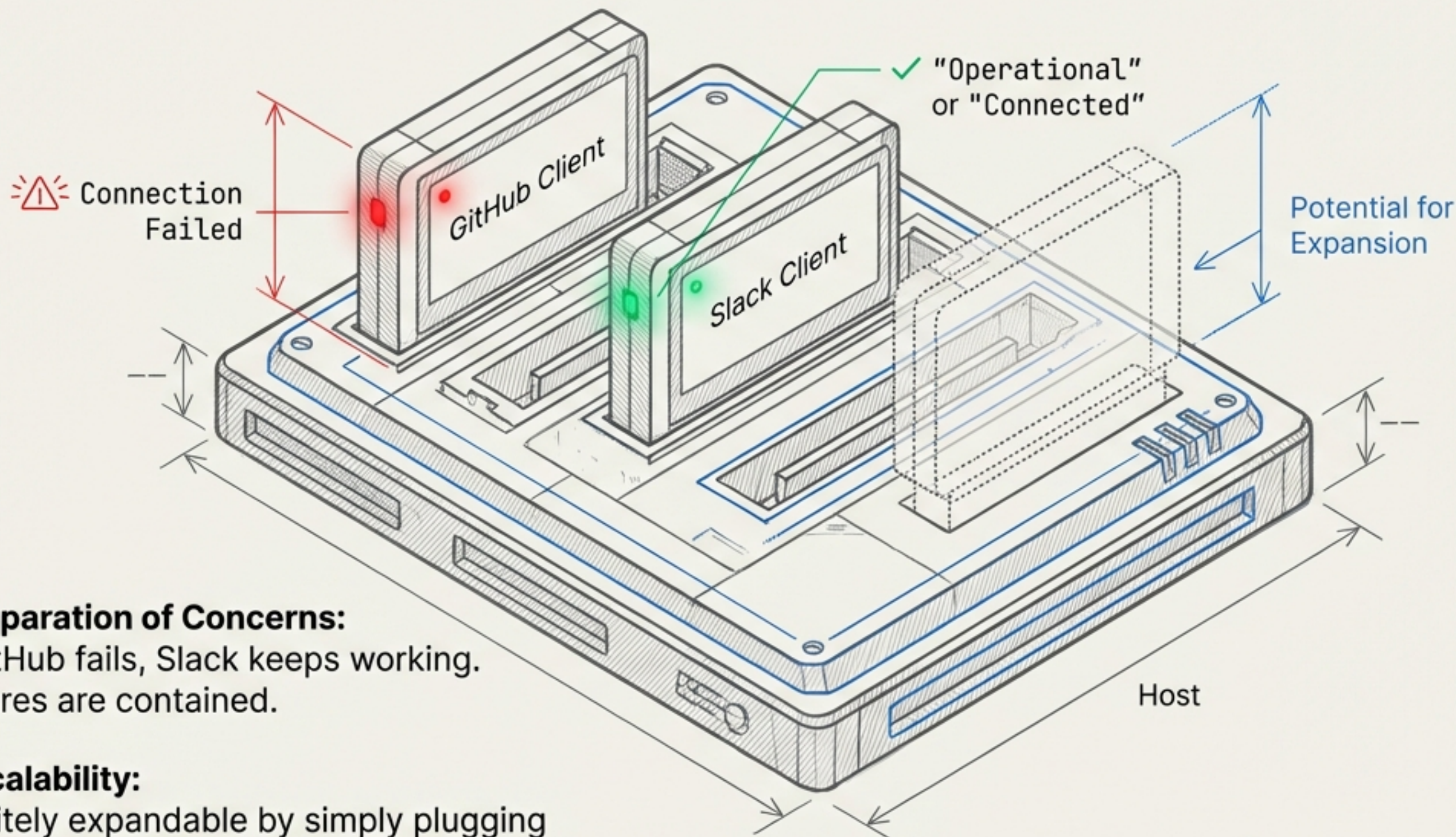
Hosts and Servers never talk directly. They need a translator.

Hosts and Servers never talk directly. They need a translator.



MODEL CONTEXT PROTOCOL

Why Decouple? Safety and Scalability



1. Separation of Concerns:

If GitHub fails, Slack keeps working. Failures are contained.

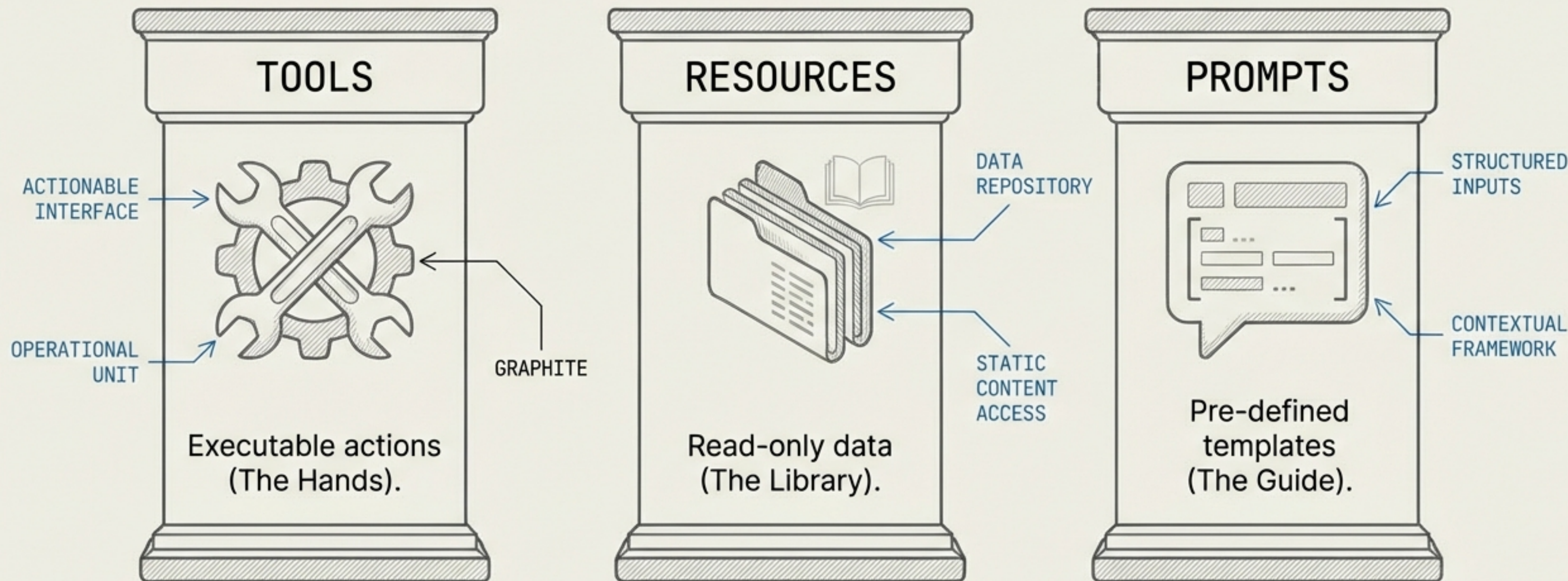


2. Scalability:

Infinitely expandable by simply plugging in new Client modules.

The Menu of Offerings: MCP Primitives

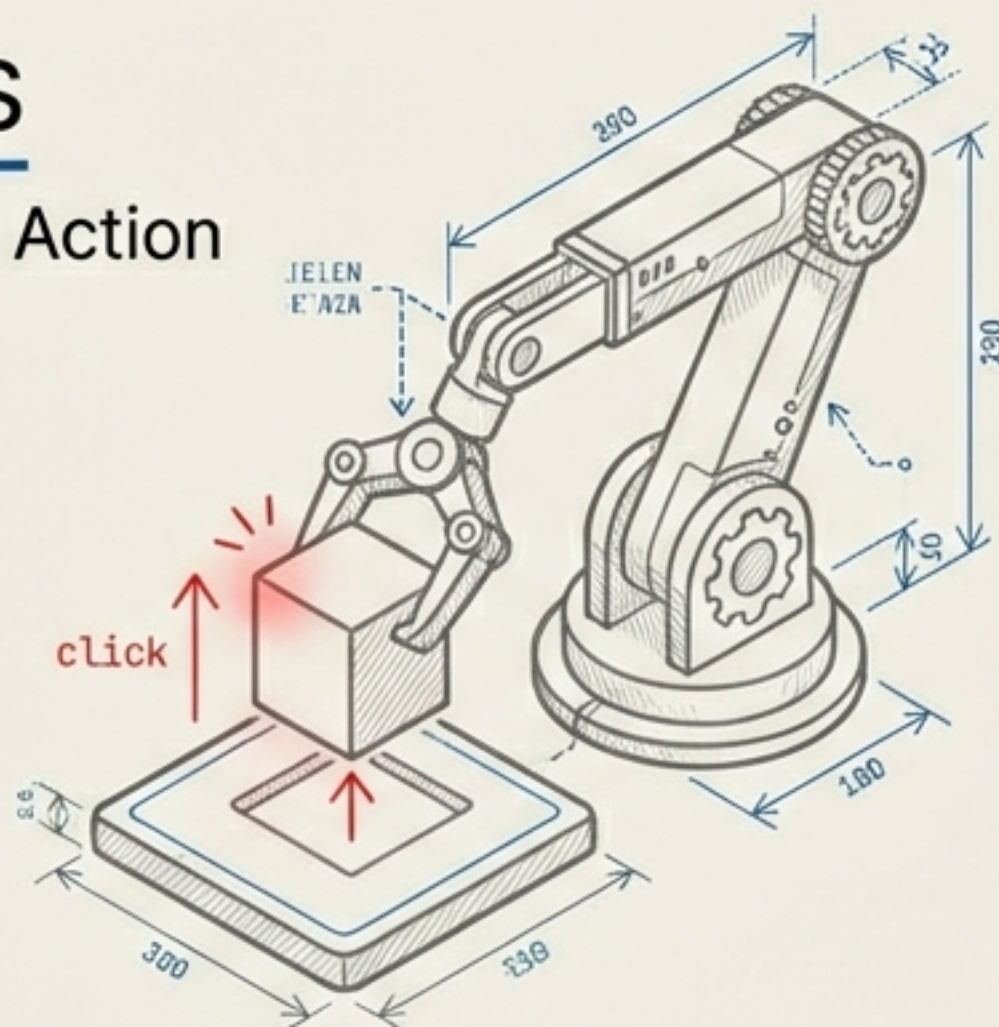
What a Server actually exposes to the Host.



Primitives Deep Dive: Doing vs. Reading

TOOLS

Dynamic Action



Dynamic / State Changing

```
call_tool("create_issue")  
call_tool("upload_file")
```

RESOURCES

Static Reference



Static / Read-Only

```
read_resource("README.md")  
read_resource("database_schema")
```


Primitives Deep Dive: Prompts

Solving vague inputs with structured templates.

BEFORE

Fix the login bug.

RESULT

Fix the login bug erroring
login bug pointle(), [unclear]
moving to, loginError is
correct. [unclear] login bug
has a undescription, [unclear].

AFTER

Use Issue Report Prompt.

ISSUE REPORT PROMPT

TITLE:

Login Error

STEPS:

1. Open App...
2. Navigate to...
3. Click Login...

ENVIRONMENT:

Chrome/Mac

RESULT

Issue #123: Login Error

Description:

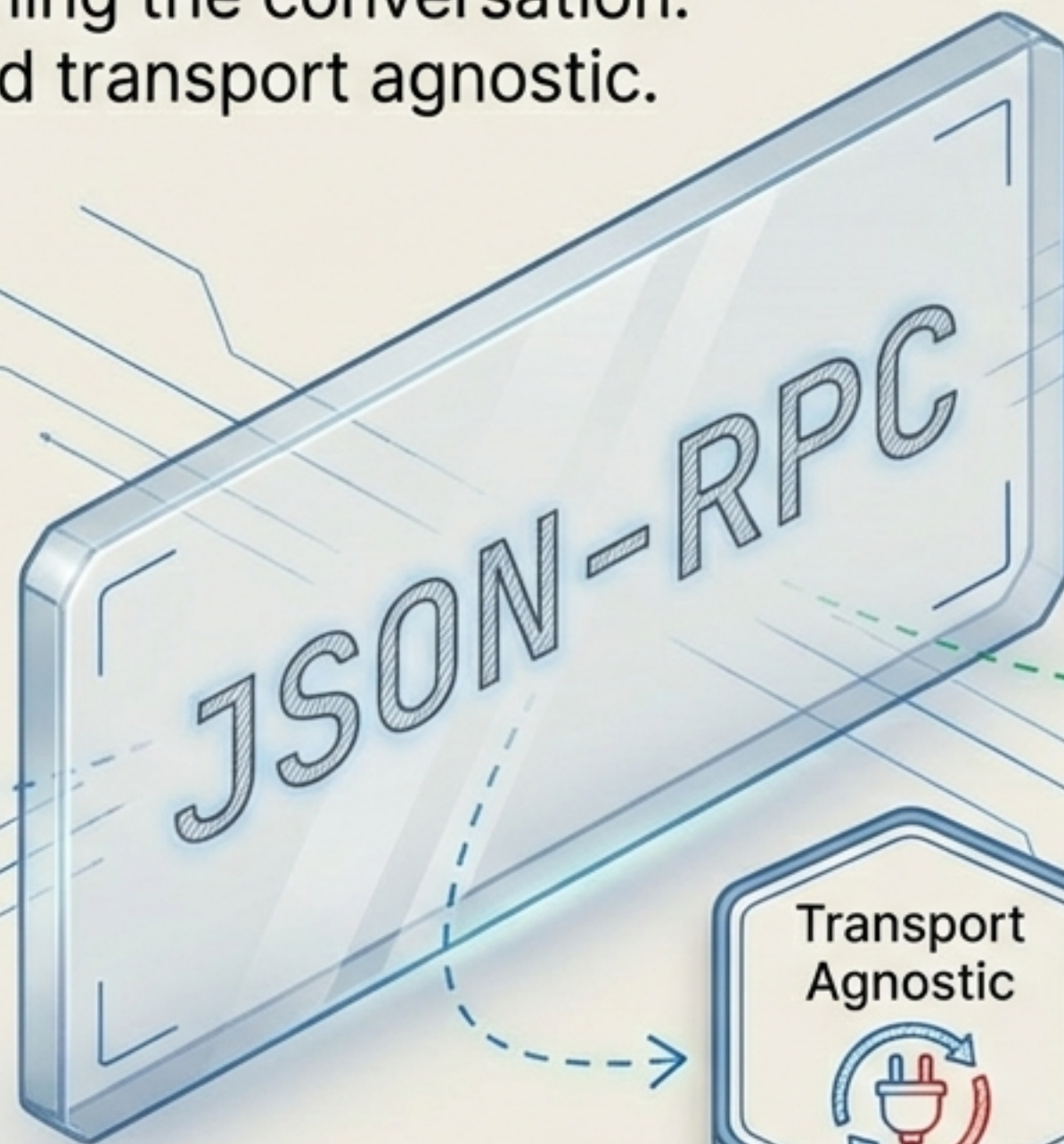
Steps to Reproduce:

Environment:

Labels: [bug, critical]

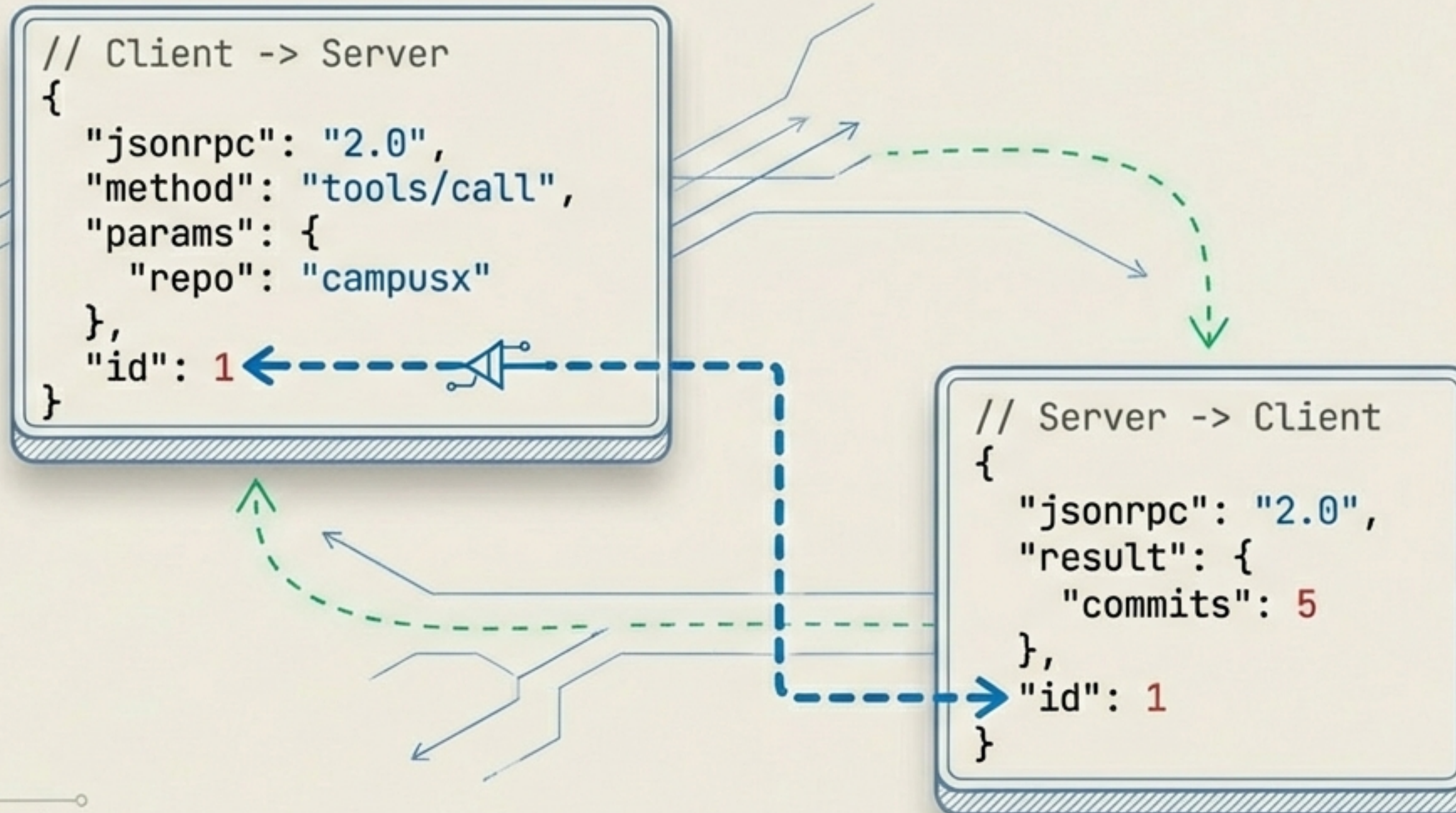
The Language: JSON-RPC 2.0

The grammatical rules governing the conversation.
Lightweight, bidirectional, and transport agnostic.



The Grammar of Conversation

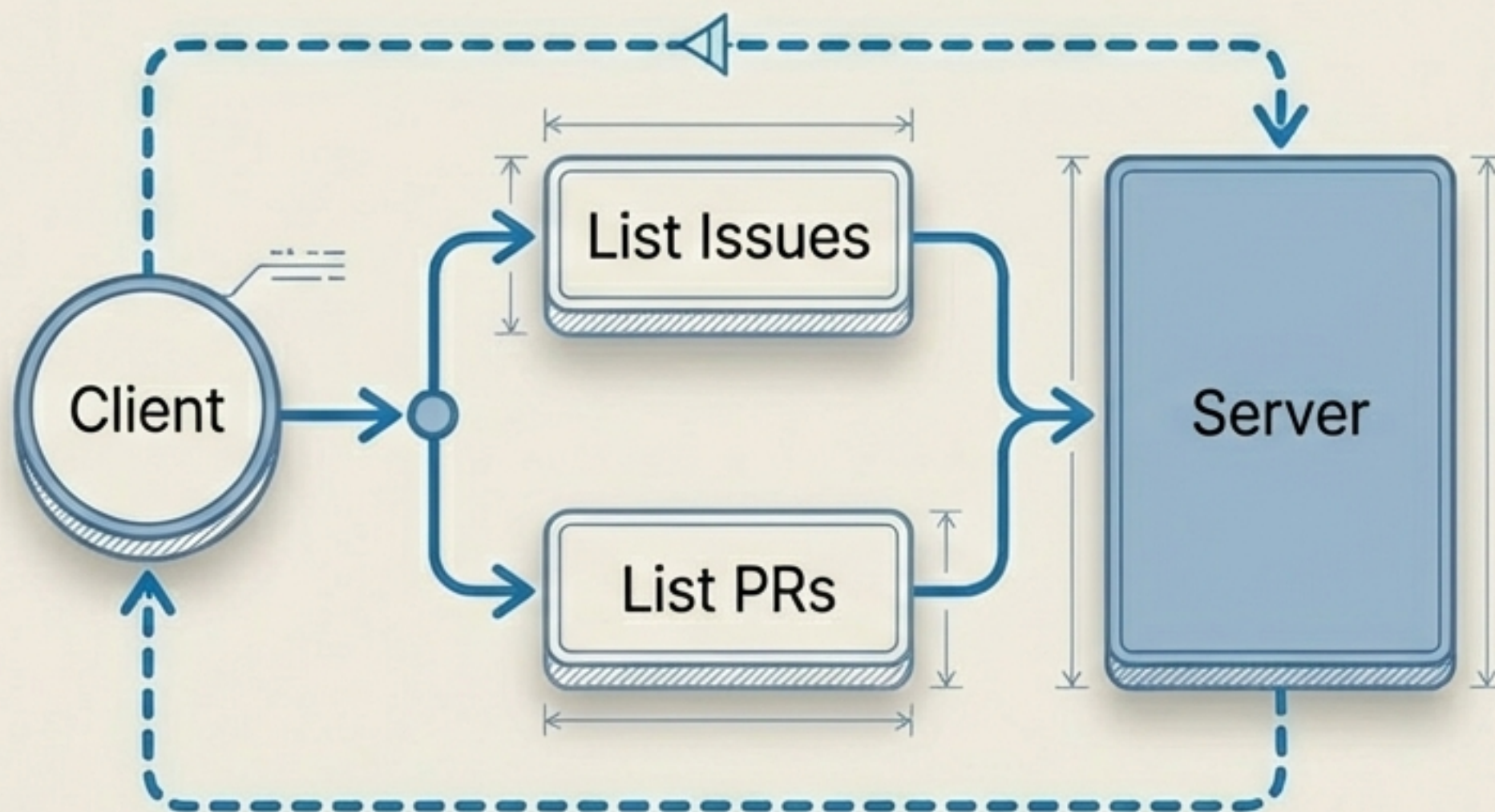
Matching Requests to Responses via ID.



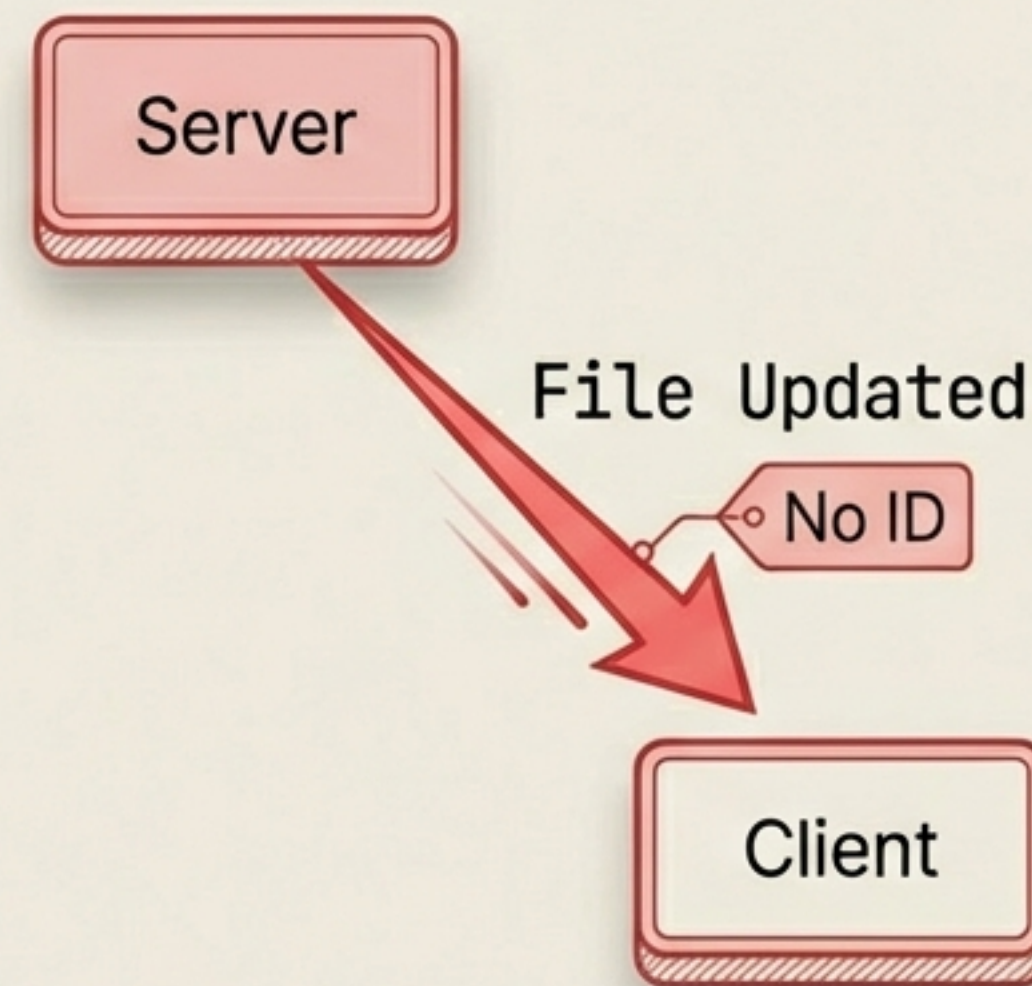
Advanced Dialects: Batching & Notifications

Optimizing communication flow for efficiency and real-time updates.

Batching

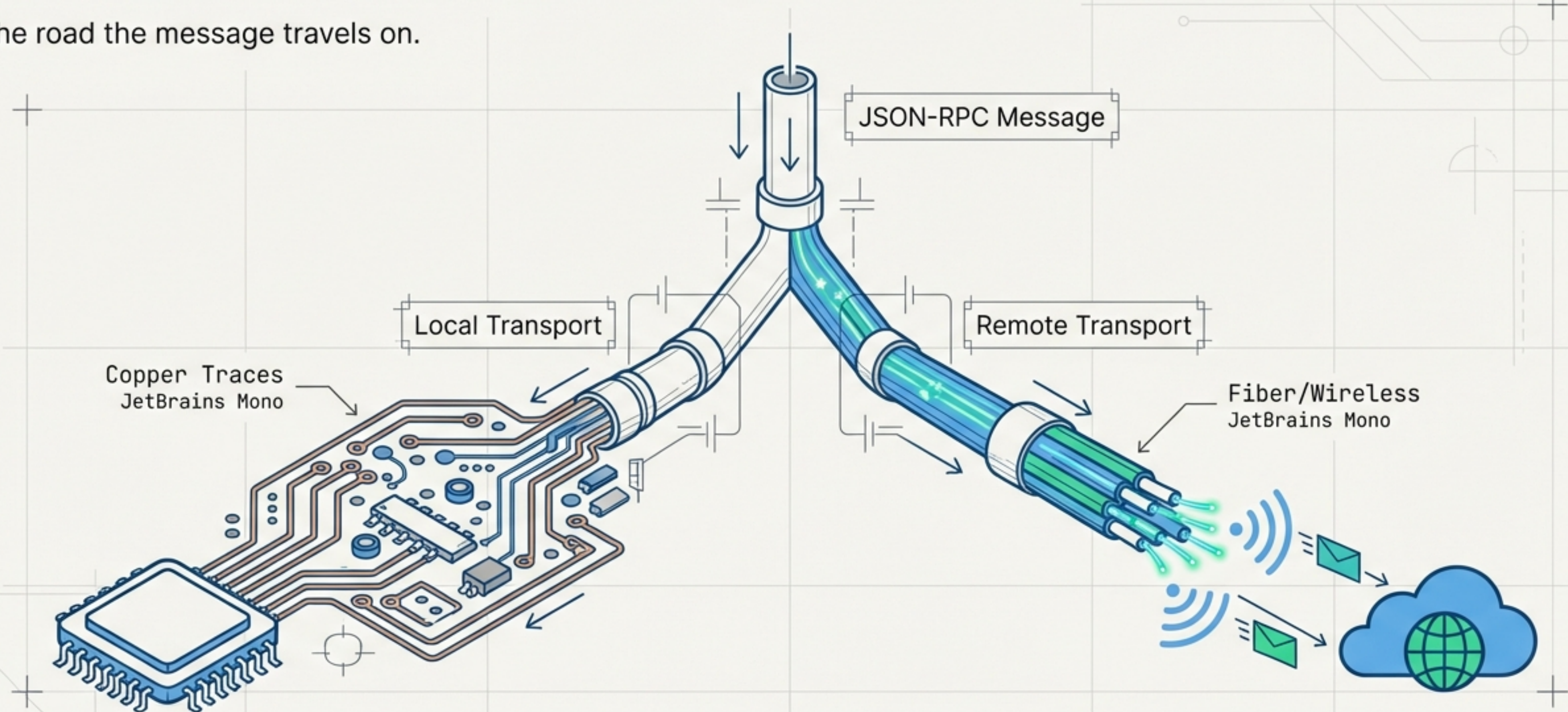


Notifications (Fire & Forget)



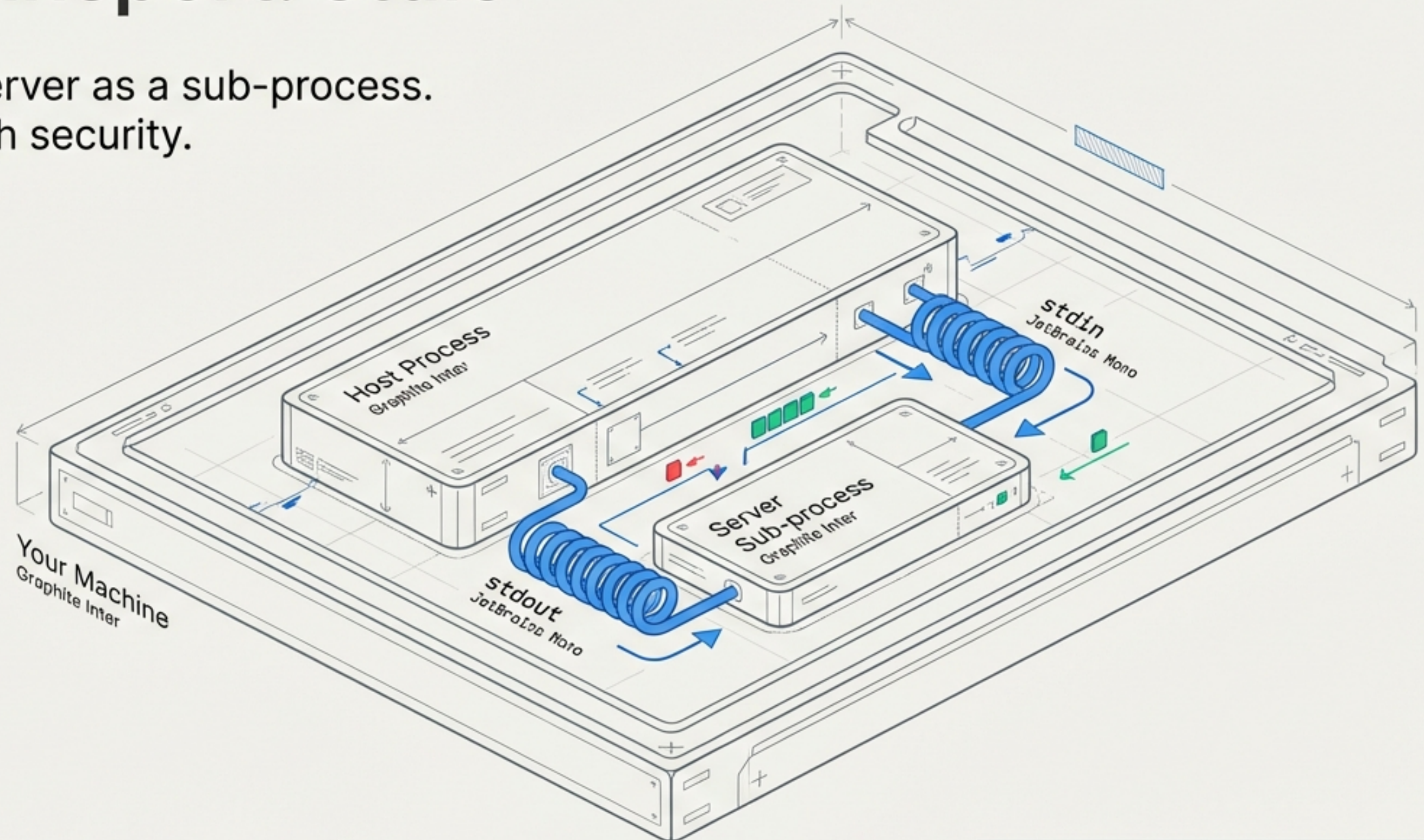
The Wires: Transport Layer

The road the message travels on.



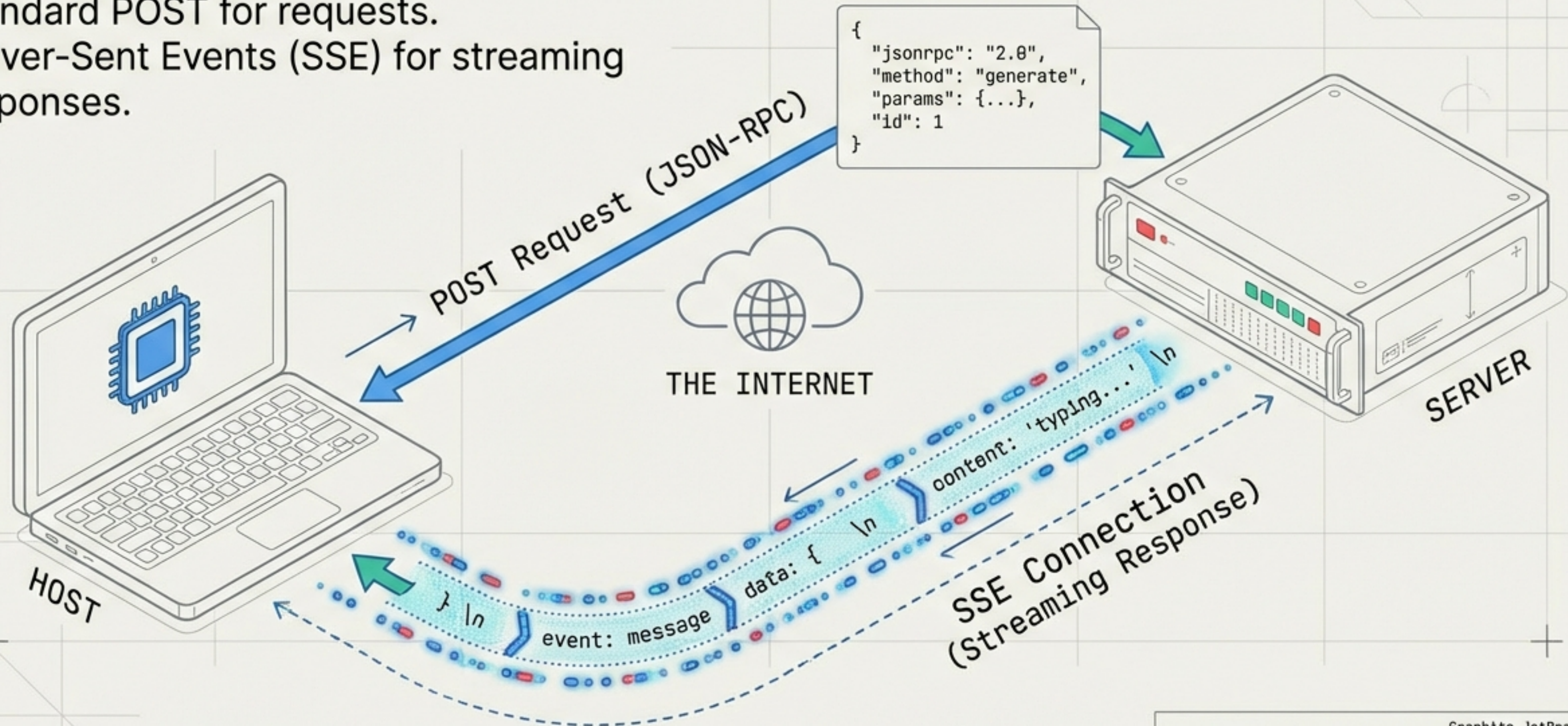
Local Transport: stdio

Host launches Server as a sub-process.
Zero latency. High security.

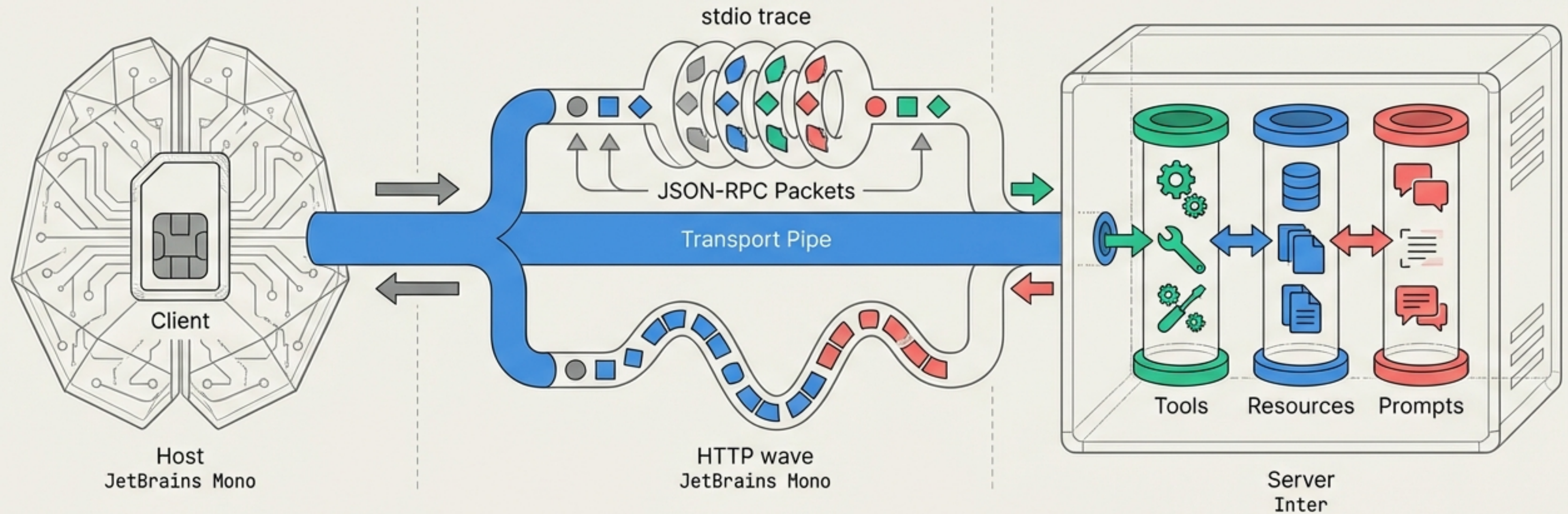


Remote Transport: HTTP + SSE

Standard POST for requests.
Server-Sent Events (SSE) for streaming responses.



The Complete Architecture



MCP decouples Intelligence from Capability, creating a modular, scalable ecosystem.